

Homework 1

Due: Friday, August 28 on Canvas

1. (5 points) A computer is programmed to reboot automatically for updates between 4:10 am and 4:55 am. The computer is equally likely to reboot anytime within this time interval.
 - (a) (2 points) What is the probability that the computer will reboot at exactly 4:16 am?
 - (b) (3 points) What is the probability that the computer will reboot between 4:15 am and 4:30 am?
2. (15 points) The sample space $S = \{(x, y): 0 \leq x \leq 3, 0 \leq y \leq 3\}$ includes values of $x, y \in \mathfrak{R}$ in the two-dimensional Cartesian space, and it includes events $\mathcal{A} = \{(x, y): y \leq 3 - x\}$ and $\mathcal{B} = \{(x, y): y \geq x\}$.
 - (a) (5 points) Draw the regions corresponding to events $\mathcal{A}$ and $\mathcal{B}$ in the (x, y) -plane.
 - (b) (5 points) Evaluate $\Pr(\mathcal{A})$, $\Pr(\mathcal{B})$ and $\Pr(\mathcal{A}|\mathcal{B})$.
 - (c) (5 points) Are $\mathcal{A}$ and $\mathcal{B}$ independent? Justify your answer.
3. (10 points) A habitually tardy person arrives at the theater late by s_i minutes, where $s_i = i, i = 1, 2, 3, \dots$. If $P[s_i] = (1/2)^i$, what is the probability that they will be more than one minute late?
4. (10 points) An urn contains 3 red balls and 2 black balls. Two balls are chosen in succession. The first ball is returned to the urn before the second ball is chosen. Each ball is chosen at random, which means that we are equally likely to choose any ball.
 - (a) (5 points) What is the probability of choosing first a red ball and then a black ball?
 - (b) (5 points) More generally, if the urn has k red balls and $N - k$ black balls. If two balls are chosen in succession and at random with replacement, what is the probability of a red ball followed by a black ball?
5. (10 points) A probability class has N students enrolled. What is the probability that at least two of the students will have the same birthday?
6. (10 points) Given a fair die, consider the following two events: A – getting 2 or 3; B – getting an even number. Are the two events independent of each other? (Hint: what are the probabilities $P[A]$ and $P[A|B]$?)
7. (10 points) Toss a fair die and consider the following three events: A – getting an even number; B – getting a number greater than 2; C – getting a number less than 5. What are the probabilities: $P[C]$, $P[B|C]$, $P[A|B, C]$ and $P[ABC]$?
8. (10 points) Suppose it is known that 0.001% of the general population has certain type of cancer. A patient visits a doctor complaining of symptoms that might indicate the presence of this cancer. The doctor performs a blood test that will confirm the cancer with the probability of 0.99 if the patient does indeed have cancer. However, the test also produces false positives or says a person has cancer when they do not. This occurs with the probability of 0.2. If the test comes back positive, what is the probability that the person has cancer?
9. (10 points) A fax machine dials a phone number that is typically busy 80% of the time. The machine dials it every 5 minutes until the line is clear and the fax is able to be transmitted. What is the probability that the fax machine will have to dial the number 9 times?
10. (10 points) A person chooses 9 letters at random from the English alphabet with replacement. What is the probability that they will be able to make the word “committee”?